

Deepraj Chouhan

AI Systems Engineer

AI Systems Engineer with 6+ years of experience building scalable backend systems, AI-powered workflow platforms, and secure cloud applications. Specialized in LLM applications, retrieval-augmented systems, AI automation, and production-grade orchestration using Python, FastAPI, OpenAI APIs, vector databases, and cloud-native infrastructure. Experienced in designing reliable AI workflows with observability, evaluation pipelines, and secure deployment architectures. UK Graduate Visa valid until Feb 2027 — no sponsorship required.

Contact



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Academic Details

- **MSc Applied Cyber Security**
University of South Wales (Sept 2023 – Sept 2024)
- **BE Information Technology SB**
Jain Institute of Technology (Aug 2016 – Aug 2020)

Soft Skills

Communication | Leadership | Problem Solving | Collaboration | Project Management | Stakeholder Engagement | Documentation

Technical Skills

Programming Languages & Frameworks:

Python | TypeScript | JavaScript | Java | SQL | C++ | React JS

AI / LLM Engineering: RAG | AI Agents | LangChain | LangGraph | OpenAI APIs | Hugging Face | Vector Databases | Fine-Tuning | Semantic Search | AI Evaluation | Retrieval Systems | LLM Observability | Agentic Workflows | AI Orchestration

Machine Learning & Data Science:

PyTorch | Scikit-learn | Pandas | NumPy | Feature Engineering

Backend & APIs:

FastAPI | Django | Node JS | Express JS | Spring Boot | REST APIs | GraphQL | WebSockets | Async Python

Cloud & Infrastructure:

AWS | Docker | Redis | CI/CD | MLflow

Databases & Storage:

PostgreSQL | MySQL | MongoDB | Firebase | Pinecone | Qdrant | S3

Profile Summary

- AI-focused software engineer with **6+ years of experience** building backend systems, AI-driven automation, and secure cloud applications.
- Experienced in developing **LLM-powered applications, retrieval-augmented generation (RAG) systems, intelligent chatbots, and workflow automation platforms** using Python, OpenAI APIs, and modern backend frameworks.
- Strong background in **full-stack development, cybersecurity, and cloud infrastructure**, enabling the design of scalable and secure AI services for real-world production environments.

Work Experience

Software Engineer

T1 Tech Dubai

Nov 2024 - Present

- **Software Engineer – React.js & .NET:** Spearhead the design and implementation of scalable product architectures, boosting system efficiency and capacity by 50%.
- **AI-Enhanced Solutions:** Integrate AI-driven automation to streamline sales processes and optimize user experiences, driving a 40% uplift in engagement and operational performance.
- **Full-Stack Development:** Designed and developed scalable backend-integrated AI-assisted applications using React, TypeScript, Python, and .NET, improving operational efficiency and automation workflows.
- **.NET & Backend Expertise:** Develop new features, troubleshoot and resolve bugs, and continuously refine existing functionality using the .NET framework, SQL Server, and Python.

Software Engineer

Seven Islands Shipping Limited

Feb 2023 – Aug 2023

- **Redesigned Architecture:** Spearheaded a comprehensive overhaul of product architecture, improving efficiency and scalability by 30%.
- **AI-Driven Enhancements:** Implemented AI-powered features for sales automation and user experience

Publications & Research

Embedding Encrypted Consent Data in Medical Images Using Advanced Steganography and Cryptography

(On going)

- Developing a secure framework combining AES-256 encryption and Pixel Value Differencing (PVD) steganography for embedding consent data in images.
- Testing robustness against compression, cropping, and steganalysis attacks to ensure data integrity and privacy compliance (HIPAA).
- Aiming to create a prototype system and technical guidelines for integration with EHR platforms in healthcare.

Core AI Systems Expertise

- Designed and deployed production-grade LLM applications using RAG pipelines, semantic retrieval, vector databases, and AI orchestration frameworks.
- Experienced in building AI workflow automation systems integrating OpenAI APIs, async processing pipelines, external APIs, and cloud-native backend architectures.
- Implemented AI evaluation and observability workflows including tracing, response monitoring, retry analysis, token usage analytics, and prompt reliability testing.
- Developed secure AI systems with prompt injection safeguards, PII masking, encrypted credential handling, access controls, and audit logging mechanisms.
- Strong understanding of retrieval systems, embeddings, reranking, contextual memory, agent orchestration, and production AI deployment tradeoffs.

optimization, resulting in a 40% improvement in engagement and operational efficiency.

- **Cross-Functional Leadership:** Led a team of four engineers while collaborating with product managers and designers to deliver high-impact solutions.
- **Frontend and Mobile Development:** Developed scalable frontend applications using React and TypeScript and launched a cross-platform mobile app broadening reach to Android and iOS.
- **Data Analytics Innovations:** Engineered a robust Python scraper for efficient website data extraction, empowering data-driven decision-making.

System Engineer

Tata Consultancy Services

Feb 2021 – Jan 2023

- **SAP CRM Security Management:** Ensured robust data security and compliance by managing the SAP CRM Security module and performing quarterly User Access validations.
- **Advanced Security Testing:** Conducted manual application security testing and grey box testing using SAP Debugger, reducing vulnerabilities and improving system resilience.
- **Cloud Migration Success:** Led the successful migration of the SAP CRM system to Oracle Cloud, ensuring seamless project execution and delivery.
- **Collaboration and Project Management:** Collaborated closely with the client team to meet project requirements and conducted regular meetings to update stakeholders, improving project transparency and efficiency.
- **Proactive Network Analysis:** Monitored network traffic and client activities on the SAP platform

Part-time Full-Stack Engineer

Fireblaze Technologies Pvt Ltd

June 2016 – Jan 2021

- **End-to-End Solution Development:** Designed and implemented the E-Learning platform 'Ignite,' boosting user engagement and functionality by 40%.
- **Website Performance Optimization:** Maintained and optimized the main website and blog platforms, improving performance, security, and user engagement by 50%.
- **Collaboration and Reporting:** Regularly reported product issues and insights to the founders and worked closely with the design and marketing teams to enhance user experience and plan effective campaigns.
- **Data-Driven Marketing:** Conducted weekly meetings with the digital marketing team, leveraging user behavior data to design and implement streamlined, efficient marketing strategies.
- **Design Challenge Resolution:** Collaborated with the design team to overcome challenges, leading to improved platform usability and higher user satisfaction.

- **Apply Forge Pro — AI Workflow Intelligence Platform**
 - Developed a production-grade AI workflow platform automating job discovery, application personalization, tracking, and recruiter outreach using LLM-powered orchestration pipelines.
 - Built retrieval-enhanced personalization systems using semantic search, embeddings, and contextual candidate memory to improve application relevance and automation accuracy.
 - Implemented AI evaluation infrastructure including response scoring, retry analysis, token analytics, and workflow observability for production monitoring and optimization.
 - Designed secure AI workflows with PII masking, audit logging, prompt injection safeguards, and encrypted credential handling.
 - Architected scalable backend services using FastAPI, PostgreSQL, Redis, vector databases, and async processing pipelines.
- **Practical Implementation and Testing of Spectre Attack**
 - Created and executed a proof of concept (POC) for the Spectre vulnerability (CVE-2017-5753 and CVE-2017-5715).
 - Demonstrated the function and impact of the Spectre exploit through detailed testing.
 - Conducted comprehensive tests and documented the outcomes to analyze the exploit's effectiveness.
 - Implemented and tested countermeasures to mitigate the Spectre attack, showcasing improved security measures.
 - Provided thorough analysis and reporting on the vulnerabilities and the effectiveness of implemented countermeasures.
- **Traffic Segregation Using Machine Learning on CICIDS2017 Dataset**
 - Utilized the CICIDS2017 dataset to identify and segregate network traffic.
 - Applied supervised and unsupervised machine learning algorithms using Python and R.
 - Conducted experiments on selected virtual machines to ensure a controlled and reproducible environment.
 - Evaluated the performance of various algorithms in differentiating between attack and normal traffic, providing detailed analysis and recommendations.